

Deep Learning

Theoretical Exercises - Lesson 3 - Chapter 4

Exercises on the book "Deep Learning" written by Ian Goodfellow,
Yoshua Bengio, and Aaron Courville.

Exercises and solutions by T. Méndez and G. Schuster

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1 Exercises on Numerical Computation

1. Given is the loss function

$$f(\mathbf{x}) = \frac{1}{4} x_1^4 + x_1^3 - \frac{17}{4} x_1^2 - 6 x_1 + \frac{1}{5} x_2^4 + \frac{6}{5} x_2^3 + 89. \quad (1.1)$$

- (a) Determine the gradient and the Hessian matrix of $f(\mathbf{x})$.
- (b) Determine all critical points and, if possible, also determine the type of the critical point (local minimum, local maximum or saddle point) using the Hessian matrix.
- (c) Determine the global minimum.
- (d) Search for a minimum by using the steepest descent method. In doing so, start at the following points and use the indicated learning rates:
 - i. Start point: $\mathbf{x}^{(0)} = [0 \ -0.5]^T$, learning rate: $\epsilon = 0.1$
 - ii. Start point: $\mathbf{x}^{(0)} = [-2.5 \ 4]^T$, learning rate: $\epsilon = 0.05$

Finish the learning algorithm after 10 iterations and check whether you have found the global minimum.

- (e) Repeat Exercise (d) with Newton's method.